Problem Feasibility Report

Group: 2
Domain: Healthcare (Mental Health)
Project Title: Early Detection of Mental Health Issues from Social Media Posts

Problem Definition:

Many people describe their distress in online communities such as reddit and other social media platforms instead of seeking actual help for their mental issues. Addressing these issues, a lot of people have worked on models that read these posts and flag signs of depression, anxiety, bipolar disorder, borderline personality disorder or suicidality. For instance, in 2025, a study [1] claimed that if we add the timestamp of the post(hour, weekday, etc) to a language model through cross modal attention, it improves the detection by a lot. The study however doesn't show that much of an improvement in accuracy and it also didn't compare against modern transformer models trained specifically on mental health text like MentalBERT [2].

Our main objective in this project is to take those posts and classify them into mental health categories and investigate two questions:
1. Do timing features genuinely improve detective beyond the text itself?
2. How well do these models show up when tested on newer posts than they were trained on?

To answer these, we will reimplement and compare 3 methods under identical conditions on public datasets:

- A TF-IDF approach with logistic regression baseline
- A fine tuned MentalRoBERTa model
- A BiLSTM with cross modal attention model from [1]

Motivation and Real-World Relevance

Around 280 million people live with depression and 301 million with anxiety disorders (WHO), and suicide caused about 727000 deaths in 2021. Suicide is a leading cause of death among 15 to 29 year olds. In low and middle income countries, more than 75% of people with mental disorders receive no treatment and people often wait years between the first symptoms and first care [3][4].

People in distress who post in online support communities, especially young people (nearly half of US teens say they are online "almost constantly") are the target audience. Also the volunteer moderators and peer supporters who must read huge volumes of posts and the counselling and crisis services that could reach people sooner are the ones we want to point out on helping towards.

When the question is “How can we help them?”, a triage tool that could bring high risk posts to trained human moderators faster with confidence and an explanation of why it was

flagged. Even though our datasets are mostly based on posts in reddit, which is not maintained under official authorities, that kind of raw data will help us train our models way nicely. It would support human judgement, not replace it or diagnose anyone. At the same time, our evaluation shows which signals can be trusted nicely, which feature set to use. It checks whether a claimed improvement is real and whether a model degrades over time, so that if communities deploy these tools, it doesn't quietly fail over time.

Proposed AI Task:

We will use supervised multi class text classification NLP techniques for this task. Our input and output is mentioned below:

Input: Reddit post (title and body) and metadata(post timestamp, etc.)
Output: Predicted category (Depression, anxiety, bipolar disorder, borderline personality disorder or suicidality) with normalized probability and a word-level explanation(in some techniques)

We will reimplement and compare three methods under identical splits and metrics (macro-F1, recall on high-risk classes, AUPRC, calibration error):

- A TF-IDF approach with logistic regression baseline
- A fine tuned MentalRoBERTa model
- A BiLSTM + LSTM with cross modal attention model over text and temporal features from [1]

Our main research question is to find out that, do temporal features significantly improve detection over a strong text only model and how much does each approach degrade when tested on posts from a later time period than it was trained on?

Challenges:

Our initial steps have been to do our literature review on different kinds of research done in this domain, where we gathered 11 papers. There are a few challenges that we have predicted as of now in our preliminary searches:

- In our anchor paper, only a small change in f1 score was recorded after using those temporal features from a single validation split. It never compared with a domain adapted transformer as well. So, it's unclear whether these features improve the model or the gain is just noise.
- Models are trained and tested on the same period, and how people talk about distress changes over time like the covid time vs now, we want to find out if that model still works for newer data properly.
- The labels for our data come from subreddit not an actual diagnosis and even though we have options for very popular data from kaggle, this might still be a question, if the data is good or not.

Feasibility Analysis:

Feasibility Check	What students should determine	Your Answers with details
Data	Do we have a suitable and accessible dataset?	YES
Research & Baselines	Are there enough recent papers and 2–3 reproducible methods?	YES
Computing & Tools	Can we run required models within resources and $0 budget?	YES
Time & Complexity	Can 5 students realistically complete Phase I and II in semester?	YES
Evaluation	Can we objectively evaluate methods and answer research question?	YES

Details:
Data: Public datasets ready for use: Kaggle Mental Disorders Reddit Dataset (~701k posts with UTC timestamps) [5] and Low et al. (2020) Zenodo dataset (3.1 GB, 28 subreddits) [6]. No data agreements required.
Research & Baselines: 5 core papers identified. 3 clear reproducible baselines
Computing & Tools: Models run on free Google Colab T4 GPU instances and we also have aws bedrock access if needed for the future timeline of the project. Classical models train in minutes and MentalRoBERTa finetunes in ~1 to 2 hours. Zero external API fees.
Time & Complexity: Clear 5 member task division: Member 1 (Data & Splits), Member 2 (Classical Baselines), Member 3 (Transformers), Member 4 (Anchor Model & Timing Ablation), Member 5 (Phase II UI & Ethics). Every member's involvement will be there for advancement of the project and tracked properly as well.
Evaluation: Quantitative metrics: Macro-F1, High-Risk Recall, AUPRC, ECE calibration error.

Preliminary Phase II Idea:
If this project needs to go into Phase II, it will be to develop an end to end human assistant for community moderators and the plan for the paper is part of Phase I, which will be an empirical or reproducibility study which does not invent a new model but compares things in controlled conditions and its reports.

References:

[1] https://arxiv.org/abs/2503.07653
[2] https://huggingface.co/mental/mental-bert-base-uncased
[3] https://www.who.int/news-room/fact-sheets/detail/mental-disorders
[4] https://www.who.int/news-room/fact-sheets/detail/suicide
[5] https://www.kaggle.com/datasets/kamaruladha/mental-disorders-identification-reddit-nlp
[6] https://zenodo.org/records/3941387